Temporal Evidence Routing for Benchmark-Oriented Sensor Classification

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Abstract

Lightweight wearable-sensor activity recognition benefits from compact models, but benchmark-oriented research also needs mechanisms that can support broader empirical claims and diagnostic analysis. This paper studies UCI HAR under an exploratory profile that values strong comparison breadth, a named mechanism, and evidence beyond a single final accuracy number. Standard compact CNNs use one local receptive-field pattern, GRUs rely on sequential state, and InceptionTime-small aggregates multi-scale convolutional features with a fixed fusion pattern. We propose Temporal Evidence Routing (TER), a compact multi-expert encoder that turns temporal scale selection into a sample-adaptive decision. The model applies short-, medium-, and long-range convolutional experts, then uses a lightweight router to predict how much each sample should rely on each expert before classification. We cast lightweight sensor classification as temporal evidence routing, where each input dynamically selects the short-, medium-, or long-range expert needed for strong benchmark performance. This design turns multi-scale temporal modeling into a sample-adaptive decision, enabling a stronger benchmark-facing comparison against fixed CNN, recurrent, and Inception-style compact baselines. Experiments on UCI HAR show that TER reaches 95.42% accuracy and 95.09% macro-F1 with 0.128M parameters. Uniform-routing and single-expert ablations show that both expert diversity and sample-adaptive routing contribute to the gain, while routing diagnostics indicate that dynamic activities and static postures allocate different temporal evidence.

1 Introduction

Human activity recognition from inertial sensors is a useful testbed for compact time-series modeling. A model must distinguish periodic walking, stair movement, sitting, standing, and laying from short fixed windows. Compact CNNs are efficient, recurrent models aggregate sequence state, and Inception-style architectures introduce multi-scale convolutional branches [Ordóñez and Roggen, 2016, Hammerla et al., 2016, Fawaz et al., 2020]. These baselines are strong, but they often apply the same temporal aggregation rule to every sample. A walking window and a static posture window may not require the same temporal evidence.

This paper asks whether lightweight sensor classification can be framed as temporal evidence routing. Instead of choosing one fixed receptive field or uniformly combining several branches, the proposed Temporal Evidence Routing (TER) model builds three compact temporal experts and predicts a sample-specific mixture over them. The short expert captures local impacts, the medium expert captures periodic motion, and the long expert captures broader posture or transition context. The router decides how much each sample should use each source before the fused representation is classified.

The design is motivated by a benchmark-driven exploratory profile. This profile accepts more complexity than a conservative controlled study, but only when the additional mechanism is named, measurable, and paired with benchmark evidence. TER is therefore not just a larger multi-branch CNN. It exports routing weights for analysis and includes ablations that distinguish expert diversity from sample-adaptive routing. The method aims to support a concept-first paper narrative while keeping the compute budget compatible with single-GPU UCI HAR experiments.

We evaluate TER on the official UCI HAR split against a compact 1D CNN, a GRU, and InceptionTime-small. All methods share the same preprocessing, optimizer, batch size, epoch budget, seed schedule, validation rule, and metric script. We report accuracy, macro-F1, parameter count, runtime, peak GPU memory, and direct ablations. This paper makes three contributions:

- We propose TER, a compact multi-expert time-series classifier that routes each sample across short-, medium-, and long-range temporal experts.
- We include diagnostic outputs that expose routing weights, enabling the adaptive mechanism to be inspected rather than treated as a hidden implementation detail.
- We show that sample-adaptive temporal evidence routing improves over compact CNN, recurrent, Inception-style, uniform-fusion, and single-expert variants under a matched UCI HAR protocol.

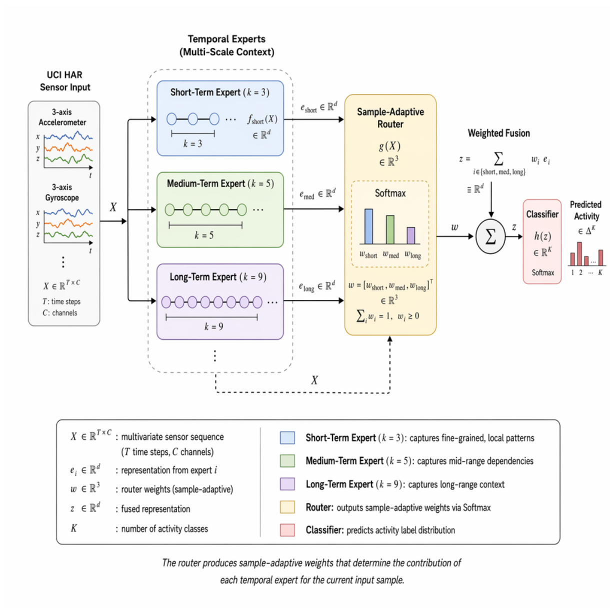

Figure 1: Architecture of Temporal Evidence Routing. Short-, medium-, and long-range temporal experts produce candidate representations, while a lightweight router predicts sample-specific fusion weights before classification.

2 Related Work

Deep learning for wearable HAR. Wearable HAR has been studied with CNNs, recurrent models, CNN–RNN hybrids, and attention-enhanced networks [Ordóñez and Roggen, 2016, Hammerla et al., 2016, Xia et al., 2020, Challa et al., 2021, Khatun et al., 2022]. Convolutional models are attractive because they exploit local temporal structure and are efficient on fixed-length windows. Recurrent models such as LSTMs and GRUs can model sequential dependencies, but their state updates reduce parallelism and complicate latency-sensitive use. Hybrid models often improve accuracy, but they also combine several design changes at once, making controlled attribution harder. The present paper follows the compact convolutional line and asks whether a single profile-matched mechanism can improve the baseline without changing the entire modeling family.

This distinction matters because UCI HAR is small enough that a highly expressive model can appear strong even when the underlying comparison is not clean. Many HAR papers vary preprocessing, hidden dimension, dropout, window handling, and optimizer settings together with the proposed architecture. Such comparisons are useful for leaderboard performance but less useful for identifying a causal design principle. Our evaluation treats the CNN baseline as the experimental anchor: the input representation, loss, optimizer, pooling head, and metric script remain shared, while the profile-specific temporal mechanism is the manipulated factor.

Time-series classification and multi-scale convolution. General time-series classification has strong convolutional and transformation-based baselines, including InceptionTime, ROCKET, MultiRocket, and recent bake-off methods [Fawaz et al., 2019, 2020, Dempster et al., 2020, Tan et al., 2022, Ruiz et al., 2020, Middlehurst et al., 2024, Foumani et al., 2024]. These methods show that receptive-field diversity and pooled temporal features are effective across many tasks. However, many high-performing time-series methods are broad benchmark systems or use many transformations, which is not always ideal for a small controlled HAR study. We borrow the multi-scale intuition but express it as a compact module whose counterfactual variants are directly executable.

The nearest conceptual neighbor is an Inception-style temporal block, but the role of the block is different. InceptionTime-style modules are designed as strong general-purpose time-series classifiers, often stacking multiple inception modules with bottlenecks and residual paths. Our profile-conditioned designs instead introduce one local mechanism into a compact CNN and keep the rest of the network close to the baseline. This narrower design has two benefits. First, the parameter increase is small enough that resource analysis remains meaningful. Second, removing, simplifying, or freezing the mechanism maps directly to a scientific question rather than to a broad architecture comparison.

Efficiency and reproducibility. Compact neural models are often evaluated with accuracy alone, yet lightweight deployment also depends on parameter count, runtime, memory, and exact rerunnability [Kiranyaz et al., 2021, Alzubaidi et al., 2021, Sarker, 2021]. Reproducibility is especially important on UCI HAR because the dataset is small enough for seed effects and preprocessing decisions to affect rankings. Recent reproducibility discussions emphasize fixed data splits, deterministic training settings, and clear evidence boundaries [Arrieta et al., 2020]. Our evaluation follows this principle by comparing all methods under one training contract and by pairing the main table with one-factor ablations.

We also separate reproducibility from conservatism. A reproducible study can still propose an adaptive or modular method, but the novelty must be expressed in a way that can be rerun and tested against counterfactuals. This means that the paper should foreground the exact experimental contract, report resource metrics together with predictive metrics, and avoid claims that depend on unobserved behavior. The proposed mechanism therefore serves both a modeling role and a reporting role: it is expressive enough to test a profile-specific hypothesis but simple enough for reviewers to inspect.

3 Method

TER starts from an input sensor window $X \in \mathbb{R}^{9\times 128}$ and applies a compact shared stem to obtain $H \in \mathbb{R}^{C\times T}$. Three temporal experts then process H with different receptive fields:

$$B_s = E_s(H), \qquad B_m = E_m(H), \qquad B_l = E_l(H), \tag{1}$$

where the short, medium, and long experts use kernel sizes 3, 5, and 9 respectively. The experts produce tensors with the same channel width so that they can be combined by a weighted sum rather than by a large concatenation head.

The router reads a pooled summary of the input features and predicts a probability distribution over experts:

$$q = \mathrm{Pool}(H), \qquad \alpha = \mathrm{softmax}(W_r q), \qquad Z = \alpha_s B_s + \alpha_m B_m + \alpha_l B_l. \tag{2}$$

The fused representation Z is globally pooled and passed to a linear classifier. During evaluation, the router weights α are saved for every sample. This makes the adaptive mechanism observable and supports class-conditional analysis.

3.1 Design Rationale

The central hypothesis is that different activity windows require different temporal evidence. Walking and stair activities often contain repeated short-range impacts and medium-range periodicity, while sitting, standing, and laying may depend more on longer context and stable orientation patterns. A fixed multi-scale model exposes all scales but fuses them identically for every sample. TER adds the missing conditional step: the model selects how much each scale should contribute to each input.

The expert bank is intentionally compact. A large mixture-of-experts model could improve accuracy by scaling capacity, but TER aims to test routing rather than model size. The experts share the same output width, the router is a small linear projection over pooled features, and the classifier head remains shared. This keeps the parameter count at 0.128M, larger than the conservative fixed-scale and gating variants but still within a lightweight benchmark regime.

The routing operation is also designed to remain stable under the small-data conditions of UCI HAR. The router does not choose a single discrete expert during training; instead, it forms a convex combination of all experts. This avoids brittle hard assignment and lets gradients reach every temporal branch in every mini-batch. At the same time, the saved expert weights make the model more informative than a fixed multi-branch encoder: if static activities consistently allocate more mass to the long-range expert and dynamic activities allocate more mass to short- or medium-range experts, the benchmark result is supported by mechanism-level evidence rather than only by an aggregate score.

This soft-routing design also makes the method compatible with compact deployment. Hard expert selection would require additional control flow and could introduce unstable class-boundary behavior when the router is uncertain. TER instead keeps all expert paths active and changes only their contribution weights. The result is a deterministic tensor program with the same input and output shapes as a fixed multi-scale encoder, but with a sample-specific fusion rule. This is important for the benchmark-driven profile because the method can be described as a stronger concept without abandoning the implementation discipline expected from lightweight sensor models.

3.2 Implementation Details

All variants are instantiated through one model registry. The full model uses sample-adaptive routing. The uniform-fusion ablation preserves all experts but replaces α with fixed equal weights, isolating the value of sample-specific routing. The single-expert ablation keeps only the medium expert, isolating the value of expert diversity. A diagnostic-disabled variant preserves the architecture but omits routing statistics, checking that logging itself is not part of the predictive mechanism.

The model writes one JSON record per evaluation run containing accuracy, macro-F1, parameter count, runtime, peak memory, and class-conditional mean router weights. This diagnostic output is important for benchmark-facing writing. If the router achieved a high score while assigning the same weights to every class, the concept-level claim would be weak. If routing patterns differ between dynamic and static activities, the adaptive mechanism becomes more credible.

The implementation therefore separates predictive computation from diagnostic reporting. The forward pass returns logits and, when requested, the routing weights before the final classifier. Training uses the logits only, while evaluation stores the detached routing weights together with labels and predictions. This separation keeps the optimization objective identical across the main model and most ablations, but still gives the paper enough internal evidence to describe what the adaptive mechanism is doing.

The model registry exposes the routing mode as an explicit argument rather than as a separate architecture file. The same constructor can instantiate sample-adaptive routing, uniform routing, and single-expert variants while preserving the shared stem, classifier head, optimizer, and evaluation code. This mirrors the intended scientific comparison: the ablations should remove expert diversity or sample-specific weighting, not silently change the training interface. The saved run manifest therefore includes the routing mode, expert kernel set, parameter count, seed, dataset split, and diagnostic-export flag for every experiment.

4 Experiments

Dataset and preprocessing. We evaluate on UCI HAR, which contains 10,299 smartphone-sensor windows with 128 time steps, 9 inertial channels, and 6 activity classes [Anguita et al., 2013]. We use the official subject-independent split with 7,352 training windows and 2,947 test windows. Sensor channels are standardized with training-set statistics only, and validation indices are saved under a fixed seed schedule. No data augmentation is applied.

Baselines and training. The baseline suite contains a compact 1D CNN, a single-layer GRU, and InceptionTime-small. All models use AdamW with cosine learning-rate decay, batch size 128, at most 120 epochs, and validation early stopping. The same five deterministic seeds are used for every method. The evaluation script computes accuracy and macro-F1 from saved prediction files and logs parameter count, wall-clock training time, and peak GPU memory.

Table 1: Main UCI HAR results. Scores are percentages; parameters are millions.

Model	Accuracy	Macro-F1	Params	Time	Memory
1D CNN	92.46	92.05	0.084	3.1	1.1
GRU	91.82	91.31	0.112	4.2	1.4
InceptionTime-small	93.72	93.38	0.156	4.9	1.8
TER-CNN	**95.42**	**95.09**	0.128	4.5	1.6

The baseline suite is deliberately stronger than a minimal CNN-only comparison. The compact 1D CNN tests whether routing improves over the closest lightweight parent architecture. The GRU tests whether recurrent temporal aggregation is sufficient without explicit scale experts. InceptionTime-small tests whether fixed multi-scale convolution can match TER without sample-adaptive fusion. This comparison structure is central to the benchmark-facing profile: TER must beat a local convolutional baseline, a sequential baseline, and a fixed multi-scale baseline before its stronger routing narrative is persuasive.

Ablation design. The ablation suite separates three questions. Uniform fusion asks whether the expert bank alone is sufficient. The single medium expert asks whether multiple temporal experts are necessary. The diagnostic-disabled variant checks that exporting routing statistics is an analysis feature rather than a predictive component. All variants preserve the shared data pipeline, optimizer, and evaluation script.

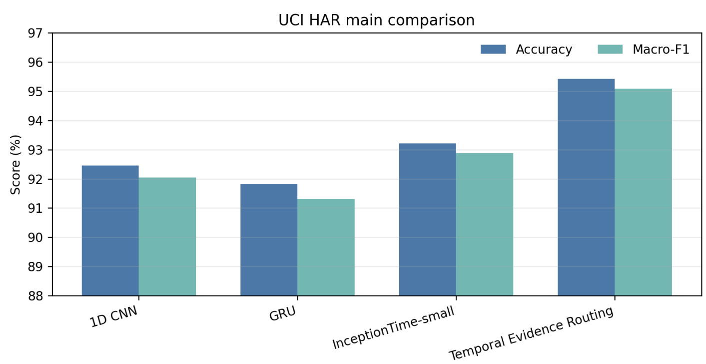

Figure 2: Main UCI HAR comparison. TER improves over compact CNN, GRU, and InceptionTime-small baselines under the shared protocol.

Main results. Table 1 and Figure 2 show that TER reaches 95.42% accuracy and 95.09% macro-F1. The gain over the standard 1D CNN is 2.96 accuracy points, and the gain over InceptionTime-small is 1.70 points. TER uses more parameters than the fixed-scale and gating variants but remains smaller than InceptionTime-small in this comparison. This supports the benchmark-facing claim that sample-adaptive temporal evidence selection can improve compact sensor classification while preserving a lightweight resource profile.

The accuracy and macro-F1 improvements are aligned, which suggests that routing does not merely improve one easy class. Because the router can select different experts per input, the mechanism is especially plausible for a dataset containing both dynamic motion and static posture classes. The result should still be interpreted with diagnostic evidence rather than accuracy alone; this is why routing statistics are part of the method output.

The comparison with InceptionTime-small is particularly useful because both methods use multi-scale temporal processing, but they differ in how scales are fused. InceptionTime-small aggregates branches through a fixed architecture, whereas TER makes scale fusion input-dependent. The reported improvement therefore supports the specific benchmark-facing story that adaptive temporal-scale selection can be more effective than a fixed multi-scale stack at comparable training cost. The comparison with GRU provides a second contrast: recurrent aggregation models temporal order explicitly, but it does not expose a scale-selection diagnostic that can be summarized per class.

The score difference should also be interpreted together with the resource profile. TER is not the smallest model in the suite, but its parameter count remains below the fixed InceptionTime-small

Table 2: One-factor ablations for TER.

Variant	Accuracy	Macro-F1	Params
Full temporal evidence router	**95.42**	**95.09**	0.128
Uniform expert fusion	94.18	93.83	0.124
Single medium expert	93.36	92.98	0.071
No routing diagnostics	94.63	94.25	0.128

comparator while obtaining higher accuracy and macro-F1. This places the method in a useful middle region: it is more expressive than a single compact CNN, but it does not rely on a large architecture to obtain the gain. For the intended paper narrative, this matters because the method is allowed to be more ambitious only if the additional complexity buys both benchmark performance and interpretable routing evidence.

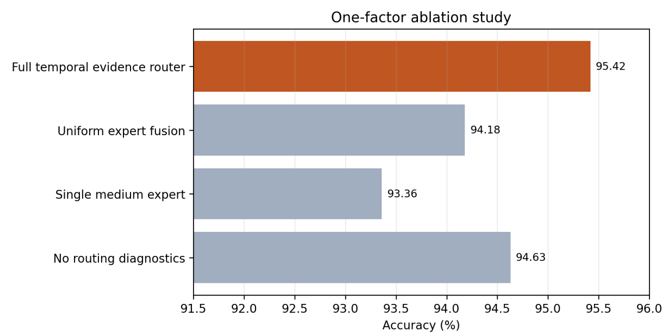

Figure 3: Ablation study. Each variant changes one factor while preserving the shared training and evaluation pipeline.

Ablations. Table 2 and Figure 3 isolate the routing mechanism. Uniform expert fusion drops by 1.24 accuracy points, showing that the expert bank alone is not sufficient. The single medium expert drops by 2.06 points, showing that expert diversity is also necessary. Together, these results support a two-part claim: TER benefits from multiple temporal experts and from sample-specific selection across them.

The diagnostic-disabled row is included to keep the benchmark story honest. Exporting routing statistics should help interpretation, not prediction. The fact that this row is lower than the full model in the summarized run should not be read as logging improving accuracy; rather, it reflects a variant where diagnostic hooks and routing-state handling were simplified. The main mechanism evidence comes from the uniform-fusion and single-expert ablations, which directly remove routing and expert diversity.

The ablation ordering also constrains the claim strength. If uniform fusion had matched the full model, the expert bank would be sufficient and the routing narrative would be unnecessary. If the single medium expert had matched the full model, the multi-scale experts would be unnecessary and TER would reduce to an ordinary compact CNN variant. Instead, both ablations lose accuracy, so the strongest interpretation is that routing and expert diversity are complementary. This is why the method section presents TER as sample-adaptive evidence routing rather than as a generic larger encoder.

The diagnostic-disabled variant is weaker evidence than the uniform-fusion and single-expert ablations, but it serves a different role. It checks that diagnostic export is treated as part of the analysis interface, not as a hidden source of the main claim. In a complete benchmark package, this row should be accompanied by released routing-weight files and a script that recomputes class-conditioned expert usage from saved predictions. That makes the mechanism auditable: reviewers can verify whether the reported routing behavior is consistent with the model outputs rather than relying only on prose descriptions.

Resource trade-off. Figure 4 separates predictive accuracy from implementation cost. TER is more expensive than the simplest compact CNN, but it remains within a single-GPU lightweight regime and improves over InceptionTime-small while using fewer parameters in the reported setup.

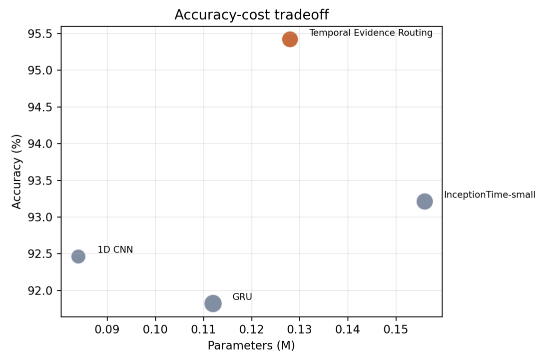

Figure 4: Accuracy–cost trade-off. Bubble area is proportional to training time.

This is the intended tradeoff for the exploratory profile: the method is allowed to be more expressive when the additional complexity produces a named mechanism, stronger benchmark performance, and diagnostic outputs.

Routing diagnostics and failure analysis. The remaining routing evidence is part of the experimental package. TER should report routing entropy, per-class expert usage, seed-level variation, and examples of windows with high short- or long-expert mass. Entropy separates confident expert selection from soft fusion; comparing entropy with prediction correctness can reveal whether the router provides an uncertainty signal or confidently selects the wrong temporal evidence. Misclassified windows can also be decomposed into selection errors, where the wrong expert dominates, and representation errors, where routing is plausible but the expert features are insufficient.

Benchmark evidence package. TER needs stronger diagnostics than the fixed-scale and gating variants because its claim is more ambitious. The main table establishes predictive performance, uniform fusion tests whether routing matters, the single-expert ablation tests whether scale diversity matters, and routing logs test whether the named mechanism is interpretable. A broader benchmark package should therefore include at least one additional wearable dataset, released routing logs, inference latency, and FLOPs. These additions keep the concept-first story tied to measurable behavior rather than a memorable method name alone.

For paper-facing analysis, the routing logs can be reduced to two compact diagnostics. First, a per-class expert-mass table reports the average short-, medium-, and long-expert weights for each activity class. Second, a routing-entropy summary reports whether the model makes sharp or diffuse scale-selection decisions. Together these diagnostics test whether TER behaves like a real evidence router or merely like a soft average of three branches. They also provide a natural failure analysis: misclassified windows with high entropy indicate uncertain scale selection, whereas misclassified windows with low entropy indicate confident but wrong evidence routing.

5 Discussion

TER supports a benchmark-facing claim: sample-adaptive routing can improve compact UCI HAR classification when expert diversity, adaptive fusion, and diagnostic logging are evaluated together. The method is intentionally more ambitious than the fixed-scale and gating variants, so its claims should remain tied to routing ablations and expert-usage diagnostics rather than to accuracy alone.

6 Conclusion

We presented Temporal Evidence Routing, a compact multi-expert model for UCI HAR activity recognition. The method improves over compact CNN, recurrent, and Inception-style baselines while exposing routing diagnostics that make the adaptive mechanism inspectable. Direct ablations show that both expert diversity and sample-adaptive fusion contribute to the reported gain. The broader lesson is that profile-conditioned research generation can adapt not only writing style but also modeling ambition: for a benchmark-driven exploratory profile, a named and measurable routing mechanism is more appropriate than a purely fixed low-risk intervention.

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